

Schrödinger Bridge for Non-linear ROMs

Reference Drifts, FFD Registration and the γ -Sweep

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Recap — Fokker–Planck Formulation

Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be the initial and terminal marginals, time rescaled to $[0, 1]$, $\gamma > 0$ the diffusion coefficient.

We seek a density–velocity pair $(\rho_t, \mathbf{b}_t)_{t \in [0, 1]}$ with $\rho_0 = \mu$, $\rho_1 = \nu$ such that

- ρ_t solves the **Fokker–Planck equation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{b}_t) = \gamma \Delta \rho_t,$$

- the pair minimizes the **expected kinetic energy**

$$\mathcal{J}(\rho, \mathbf{b}) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} |\mathbf{b}_t(\mathbf{x})|^2 \rho_t(\mathbf{x}) \, d\mathbf{x} \, dt.$$

Girsanov applied to $dX_t = \mathbf{b}_t(X_t) \, dt + \sqrt{2\gamma} \, dW_t$ gives $\text{KL}(\mathbb{P} \parallel \mathbb{R}) = \mathcal{J}/(2\gamma)$: the Schrödinger bridge is the minimal-energy stochastic interpolation between μ and ν .

Recap — Heat and Advection–Diffusion Semigroups

Hopf–Cole ($\eta = e^f$, $\eta^* = e^g$) linearizes the optimality system into two transport–diffusion PDEs, whose transition operators are the SB kernel.

Driftless bridge ($\mathbf{b} \equiv 0$) — heat semigroup $\mathcal{P}_\tau = e^{\tau\gamma\Delta}$

$$\partial_\tau \phi = \gamma \Delta \phi, \quad \partial_n \phi|_{\partial\Omega} = 0 \quad (\text{Neumann, no-flux}).$$

Drifted bridge — advection–diffusion semigroups $\mathcal{Q}_\tau = e^{\tau\mathcal{L}}$, $\mathcal{Q}_\tau^\dagger = e^{\tau\mathcal{L}^*}$

$$\underbrace{\partial_\tau \phi = \mathbf{b} \cdot \nabla \phi + \gamma \Delta \phi}_{\mathcal{Q}: \text{ non-conservative, backward}}, \quad \underbrace{\partial_\tau \phi = -\nabla \cdot (\mathbf{b}\phi) + \gamma \Delta \phi}_{\mathcal{Q}^\dagger: \text{ conservative, forward}}.$$

IPFP in log-potential form:

$$f^{(k+1)} = \log \mu - \log \mathcal{Q}_1[e^{g^{(k)}}], \quad g^{(k+1)} = \log \nu - \log \mathcal{Q}_1^\dagger[e^{f^{(k+1)}}].$$

FVM: TPFA diffusion + first-order upwind advection, SSP-RK2, Anderson-accelerated.

$$\mathbf{b} \equiv 0 \Rightarrow \mathcal{Q}_1 = \mathcal{Q}_1^\dagger = \mathcal{P}_1.$$

Recap — Convex Displacement Interpolation (CDI)

The maps act on the shock-feature marginals; the physical Mach field is recovered gradient-free. For $t \in (0, 1)$,

$$\hat{M}(t, \mathbf{x}) = (1 - t) M_0((1 - t) \mathbf{x} + t S(\mathbf{x})) + t M_1(t \mathbf{x} + (1 - t) T(\mathbf{x}))$$

- M_0, M_1 = endpoint Mach fields, sampled at the displaced pre-images by linear interpolation on the mesh;
- exact at the endpoints: $\hat{M}(0, \cdot) = M_0$, $\hat{M}(1, \cdot) = M_1$;
- no map inversion — forward pre-image $t\mathbf{x} + (1 - t)T$ pulls back M_1 , backward pre-image $(1 - t)\mathbf{x} + tS$ pulls back M_0 , blended with weights $(1 - t, t)$.

The reference drift is designed to keep T, S close to the identity, so the CDI reconstruction stays well-conditioned at small γ .

Recap — Construction of the Transport Maps T, S

T is the conditional mean of the target position given the source (built from g through Q_1); S is its backward analogue (from f through $Q_1^\dagger$):

$$T(\mathbf{x}) = \frac{Q_1[\text{id } e^g](\mathbf{x})}{Q_1[e^g](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}), \quad S(\mathbf{x}) = \frac{Q_1^\dagger[\text{id } e^f](\mathbf{x})}{Q_1^\dagger[e^f](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}).$$

Neumann identity-leak correction

$$\beta_{\text{bias}} = \frac{Q_1[\text{id}]}{Q_1[1]} - \Phi_{\mathbf{b}},$$

measured against the drift's own *reflected deterministic flow* $\Phi_{\mathbf{b}}$ (identity when $\mathbf{b} \equiv 0$), so $\mathbf{b}$'s transport stays in T, S .

Evaluated in the stable log-domain with a positivity shift $\text{id} \mapsto \text{id} - c$; a trust gate reverts kernel-collapse outlier cells to the identity. $T, S \rightarrow \text{id}$ far from the transported feature.

Oblique Drift (1/2) — Wave Angles

The LE and TE shocks rotate between their M_0 and M_1 wave angles:

θ - β - M relation (weak root) — LE family

$$\tan \theta = 2 \cot \beta_s \frac{M^2 \sin^2 \beta_s - 1}{M^2(\gamma_g + \cos 2\beta_s) + 2}, \quad \beta_s^{\text{LE}}(M) = \beta_s(\alpha, M).$$

TE family — LE shock $\rightarrow$ Prandtl-Meyer(2α) $\rightarrow$ TE shock

$$\nu(M_b) = \nu(M_a) + 2\alpha, \quad \beta_s^{\text{TE}}(M) = \beta_s(\alpha, M_b(M)).$$

Ray angles and rotation rates, $M(t) = (1 - t)M_0 + tM_1$

$$\psi_{\text{LE}} = \beta_s^{\text{LE}}, \quad \psi_{\text{TE}} = \beta_s^{\text{TE}} - \alpha, \quad \omega_\kappa(t) = \left. \frac{d\beta_s^\kappa}{dM} \right|_{M(t)} (M_1 - M_0).$$

Oblique Drift (2/2) — Windowed Rigid Rotation

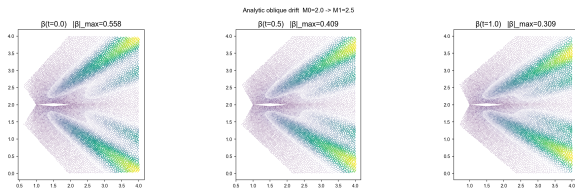
Four windowed rigid rotations about the apexes $\mathbf{a}_\kappa$

$$\mathbf{b}(t, \mathbf{x}) = \sum_{\kappa \in \{\text{LE}, \text{TE}\}} \sum_{\sigma = \pm 1} \sigma \omega_\kappa(t) w_{\kappa, \sigma}(t, \mathbf{x}) \mathbf{R}(\mathbf{x} - \mathbf{a}_\kappa), \quad \mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

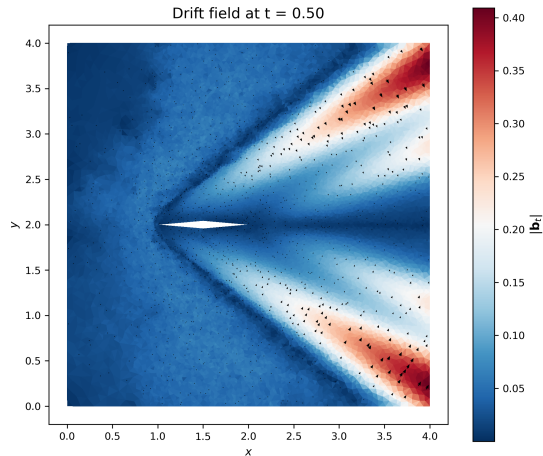
$$w_{\kappa, \sigma} = e^{-\text{dist}_{\kappa, \sigma}^2 / \sigma_w(s)^2} \mathbf{1}[s_{\kappa, \sigma} > 0], \quad s_{\kappa, \sigma} = (\mathbf{x} - \mathbf{a}_\kappa) \cdot \hat{\mathbf{d}}_{\kappa, \sigma}.$$

Zero in the freestream; the conical band $\sigma_w(s)$ covers the swept wedge; $\nabla \cdot \mathbf{b} \neq 0 \Rightarrow$ enters through the conservative $Q^\dagger$.

Oblique Drift — Field

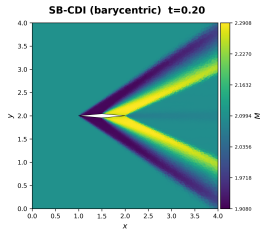


Analytic $\beta(t=0.5)$

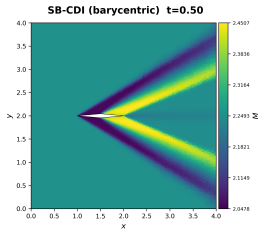


$$\text{Total drift } \mathbf{b}_t = \boldsymbol{\beta} + 2\gamma \nabla g_t$$

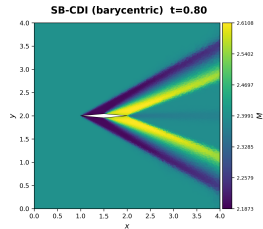
Oblique Drift — Interpolation vs. HR Reference (eig, $h = 0.025$)



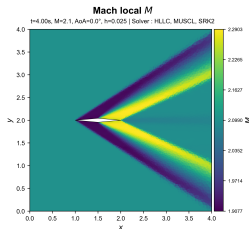
SB-CDI, $t = 0.20$



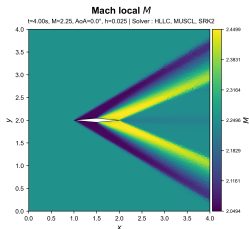
SB-CDI, $t = 0.50$



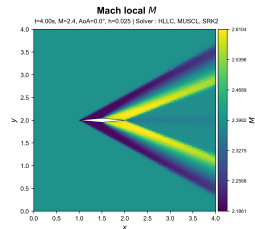
SB-CDI, $t = 0.80$



HR ref., $M = 2.10$

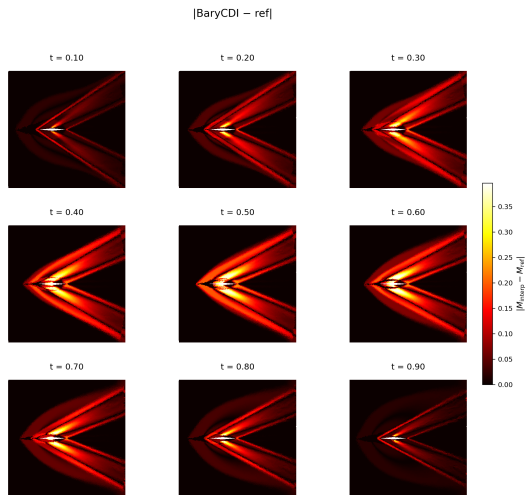


HR ref., $M = 2.25$

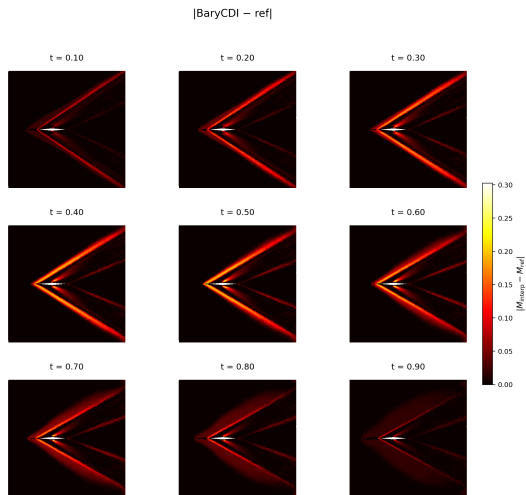


HR ref., $M = 2.40$

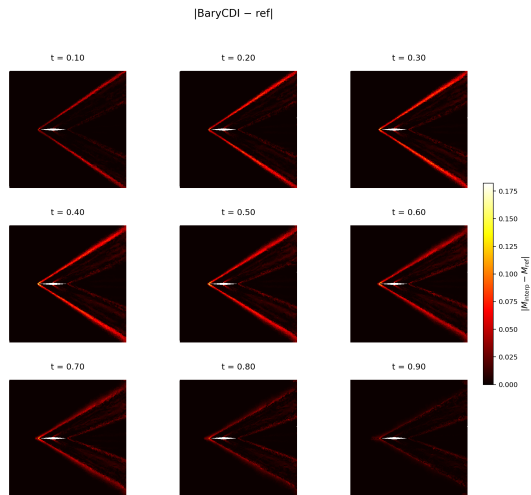
Oblique Drift — Absolute Error, $\gamma_{\min} = 0.5$



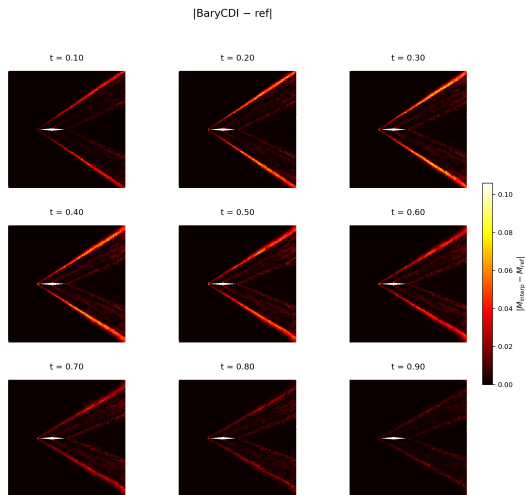
Oblique Drift — Absolute Error, $\gamma_{\min} = 0.05$



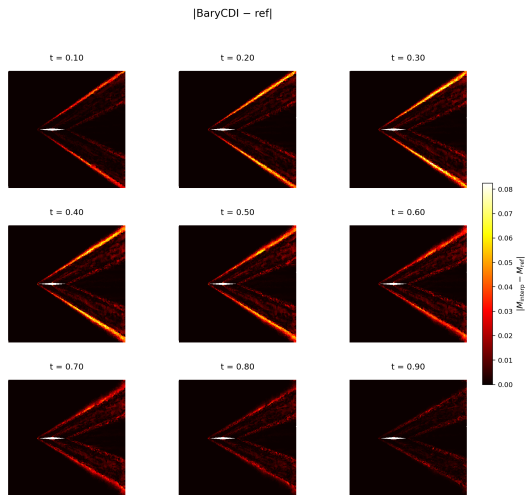
Oblique Drift — Absolute Error, $\gamma_{\min} = 0.005$



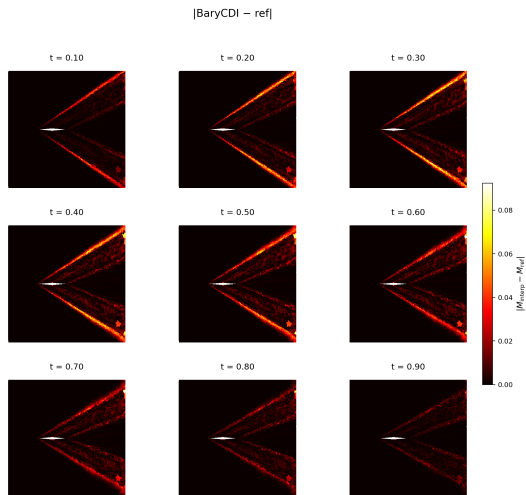
Oblique Drift — Absolute Error, $\gamma_{\min} = 5 \times 10^{-4}$



Oblique Drift — Absolute Error, $\gamma_{\min} = 10^{-4}$



Oblique Drift — Absolute Error, $\gamma_{\min} = 10^{-5}$



Free-Form Deformation (FFD)

$$T : \mathbb{R}^{N_{cp}} \times \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$\lambda, \xi \longmapsto T_{\lambda_0}(\xi) = x$$

In the coordinate system $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$

$$\mathbf{x} = \mathbf{x}_0 + t_1 \mathbf{e}_1 + t_2 \mathbf{e}_2 + t_3 \mathbf{e}_3$$

The mapping is defined as:

$$FFD(t_1, t_2, t_3) = \sum_{i=0}^l \sum_{j=0}^m \sum_{k=0}^n B_i^l(t_1) B_j^m(t_2) B_k^n(t_3) P_{i,j,k} \quad (1)$$

Here, $P_{i,j,k}$ are the control points in the deformed lattice, and l, m, n denote the degree of the polynomial in each direction.

The (t_1, t_2, t_3) coordinates can be calculated using:

$$t_1 = \frac{(\mathbf{e}_2 \times \mathbf{e}_3) \cdot (X - X_0)}{(\mathbf{e}_2 \times \mathbf{e}_3) \cdot \mathbf{e}_1} \quad (2)$$

$$t_2 = \frac{(\mathbf{e}_1 \times \mathbf{e}_3) \cdot (X - X_0)}{(\mathbf{e}_1 \times \mathbf{e}_3) \cdot \mathbf{e}_2} \quad (3)$$

$$t_3 = \frac{(\mathbf{e}_1 \times \mathbf{e}_2) \cdot (X - X_0)}{(\mathbf{e}_1 \times \mathbf{e}_2) \cdot \mathbf{e}_3} \quad (4)$$

Where a Bernstein polynomial is defined as:

$$B_i^l(t) = \binom{l}{i} t^i (1-t)^{l-i} \quad (5)$$

Data-Driven Drift — FFD Registration

Register the two shock-feature marginals u_0, u_1 by two FFD maps T (forward) and W (backward), whose control-point displacements $\Delta P^T, \Delta P^W$ are the free DOF (single region, all control points free, no boundary conditions).

Objective (single region), weights $(\alpha_A, \alpha_B, \alpha_C, \alpha_D, \alpha_G) = (1, 0.1, 1, 10^{-2}, 0.1)$

$$\begin{aligned} J = & \alpha_A (\|u_1 \circ T - u_0\|^2 + \|u_0 \circ W - u_1\|^2) \\ & + \alpha_B (\|W \circ T - \text{id}\|^2 + \|T \circ W - \text{id}\|^2) \\ & + \alpha_C (\|b_\varepsilon(\det \nabla T)\|^2 + \|b_\varepsilon(\det \nabla W)\|^2) + \frac{\alpha_D}{N_{cp}} (\|\Delta P^T\|^2 + \|\Delta P^W\|^2) \\ & + \alpha_G (\mathcal{W}_2^2(u_1 \circ T, u_0) + \mathcal{W}_2^2(u_0 \circ W, u_1)) \end{aligned}$$

α_A alignment (both directions) · α_B bijectivity / map-sync · α_C Jacobian log-barrier (positivity) · α_D control-point reg. · α_G $\mathcal{W}_2$ capture.

T, W analytic (Bernstein) $\Rightarrow$ evaluated *exactly* at the true cell centres — no grid $\rightarrow$ mesh interpolation error.

From FFD Mappings to the Reference Drift β

The alignment term drives $u_1(T(\mathbf{x})) = u_0(\mathbf{x})$: at a field-0 feature p_0 , $T(p_0) = p_1$ (its field-1 position). Evaluating the converged Bernstein maps at the true cell centres and taking the frozen unit-scale ($s=1$) displacements,

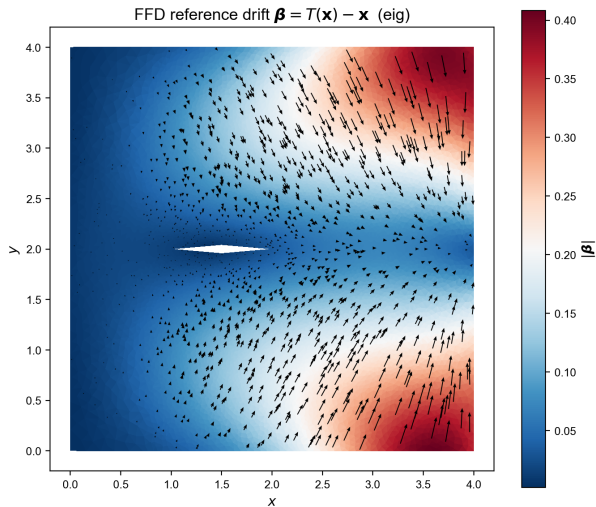
$$\beta(\mathbf{x}) = T(\mathbf{x}, s=1) - \mathbf{x}$$

$$\alpha(\mathbf{x}) = W(\mathbf{x}, s=1) - \mathbf{x}$$

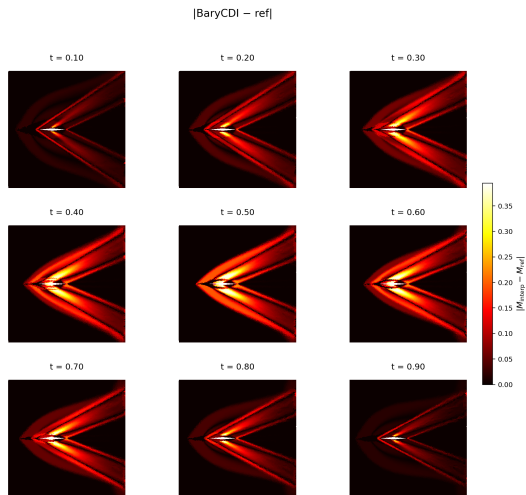
forward ($\mu_0 \rightarrow \mu_1$) via β ; backward ($W \approx T^{-1}$) via α .

- β is a frozen $(N, 2)$ field, consumed by the SB kernel $\mathcal{Q}/\mathcal{Q}^\dagger$, $\Phi_{\mathbf{b}}$ -referenced bias, trust gate and CDI *exactly* like the oblique drift;
- β is cached per marginal pair (`ffd_beta_⟨hash⟩.npz`) — the registration is offline, run once;
- the raw maps $T = \mathbf{x} + \beta$, $S = \mathbf{x} + \alpha$ also serve as a *stand-alone* interpolator (FFD control).

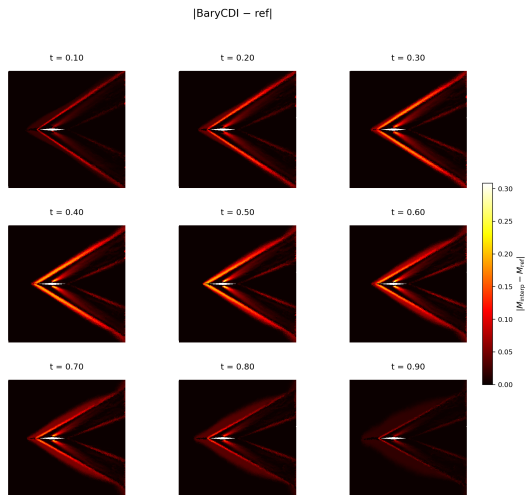
FFD Reference Drift $\beta = T(\mathbf{x}) - \mathbf{x}$



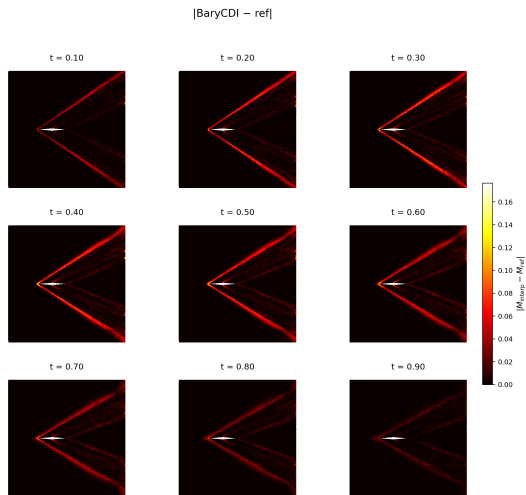
FFD Drift — Absolute Error, $\gamma_{\min} = 0.5$



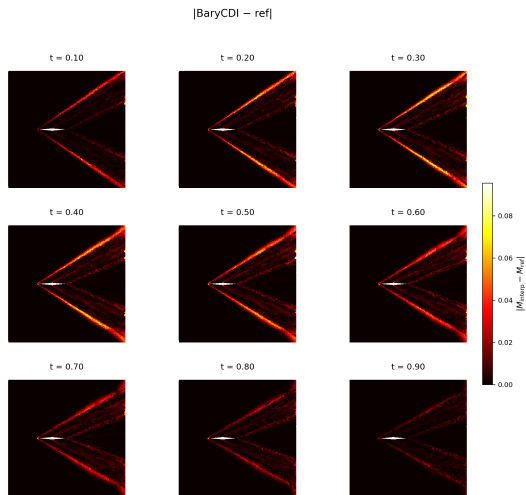
FFD Drift — Absolute Error, $\gamma_{\min} = 0.05$



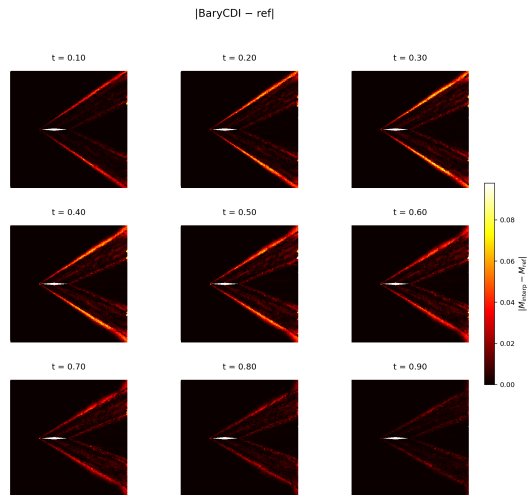
FFD Drift — Absolute Error, $\gamma_{\min} = 0.005$



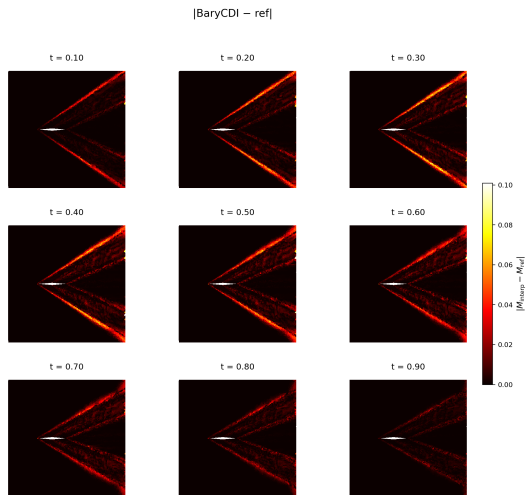
FFD Drift — Absolute Error, $\gamma_{\min} = 5 \times 10^{-4}$



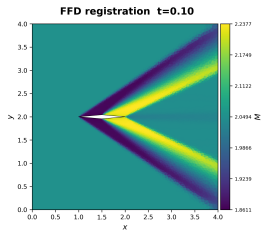
FFD Drift — Absolute Error, $\gamma_{\min} = 10^{-4}$



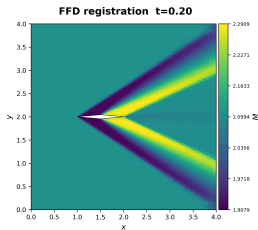
FFD Drift — Absolute Error, $\gamma_{\min} = 10^{-5}$



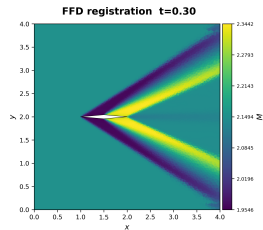
FFD Registration as Interpolator vs. HR Reference (1/3)



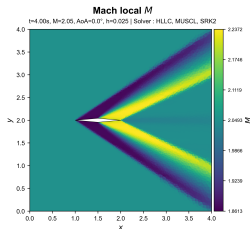
FFD, $t = 0.10$



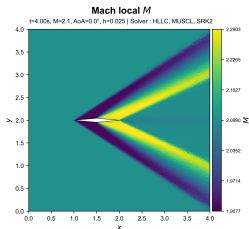
FFD, $t = 0.20$



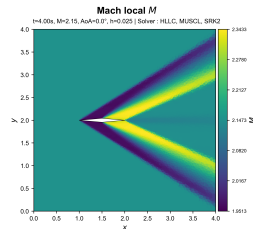
FFD, $t = 0.30$



HR ref., $M = 2.05$

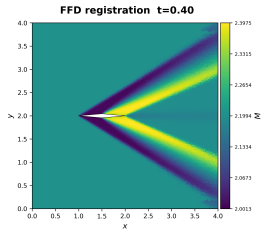


HR ref., $M = 2.10$

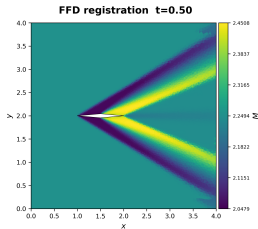


HR ref., $M = 2.15$

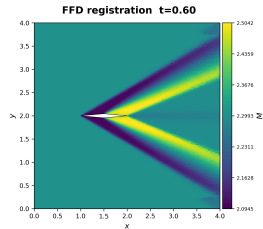
FFD Registration as Interpolator vs. HR Reference (2/3)



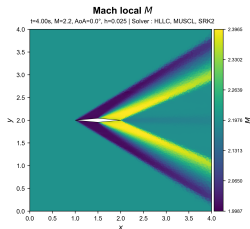
FFD, $t = 0.40$



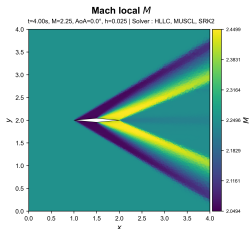
FFD, $t = 0.50$



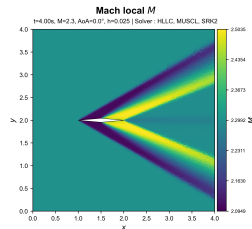
FFD, $t = 0.60$



HR ref., $M = 2.20$

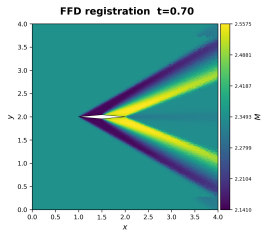


HR ref., $M = 2.25$

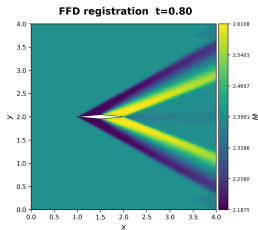


HR ref., $M = 2.30$

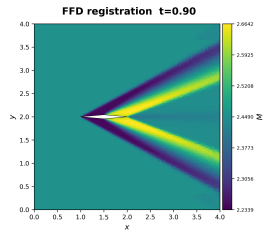
FFD Registration as Interpolator vs. HR Reference (3/3)



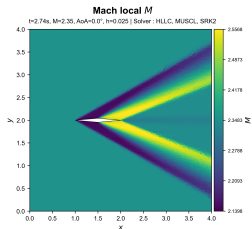
FFD, $t = 0.70$



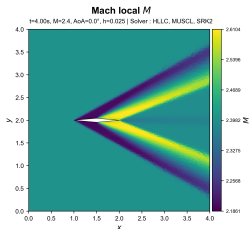
FFD, $t = 0.80$



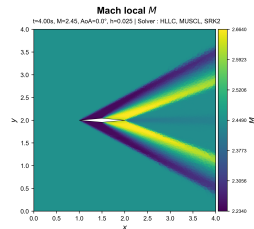
FFD, $t = 0.90$



HR ref., $M = 2.35$

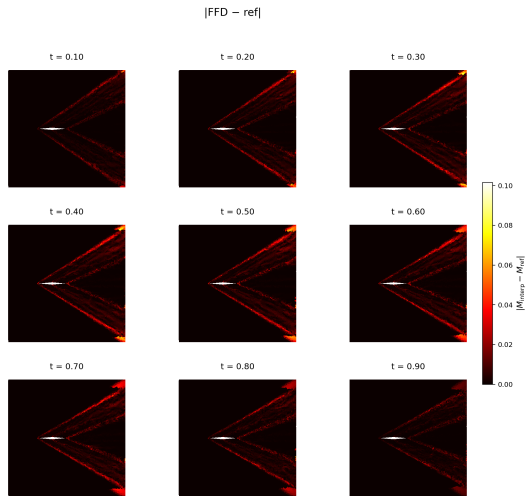


HR ref., $M = 2.40$



HR ref., $M = 2.45$

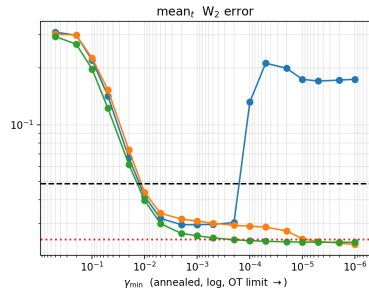
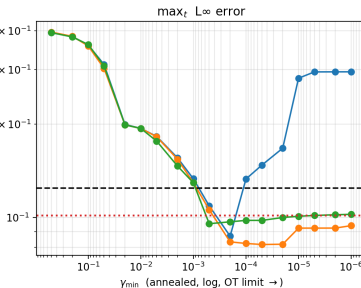
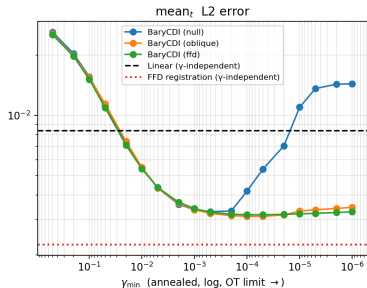
FFD Registration as Interpolator — Absolute Error



γ -independent (raw registration maps $T = \mathbf{x} + \beta$, $S = \mathbf{x} + \alpha$).

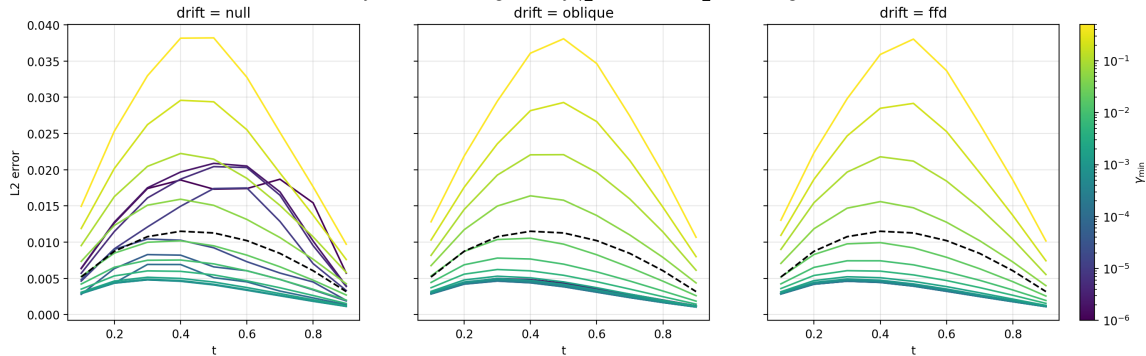
Sweep — Error vs. $\gamma_{\min}$ (eig)

Interpolation error vs. annealed $\gamma_{\min}$ — hessian_mode = eig



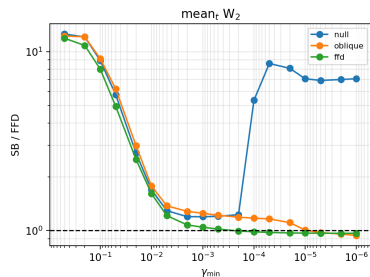
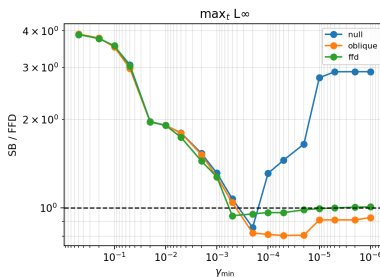
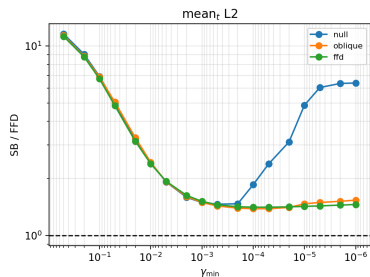
Sweep — Error vs. t (eig)

BaryCDI L2(t) colour-graded by $\gamma_{\min}$ — hessian_mode = eig

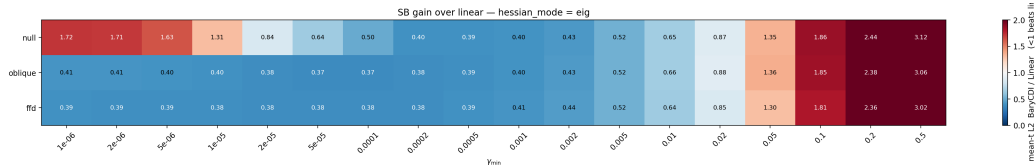


Sweep — SB vs. FFD Registration (eig)

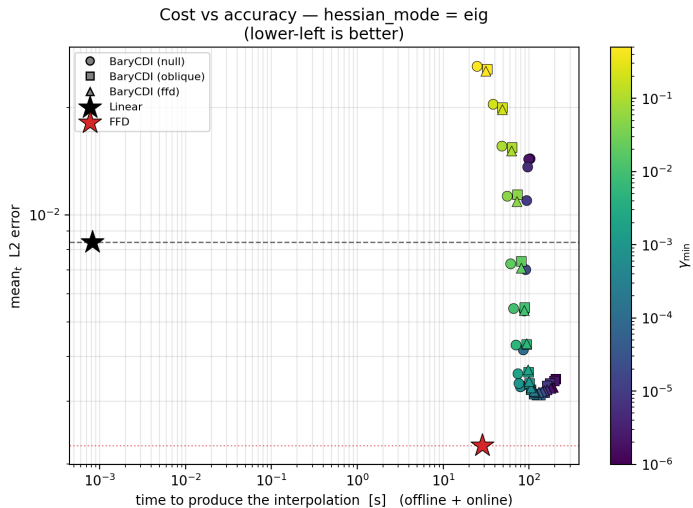
Schrödinger bridge vs. raw FFD registration (<1 = SB better) — hessian_mode = eig



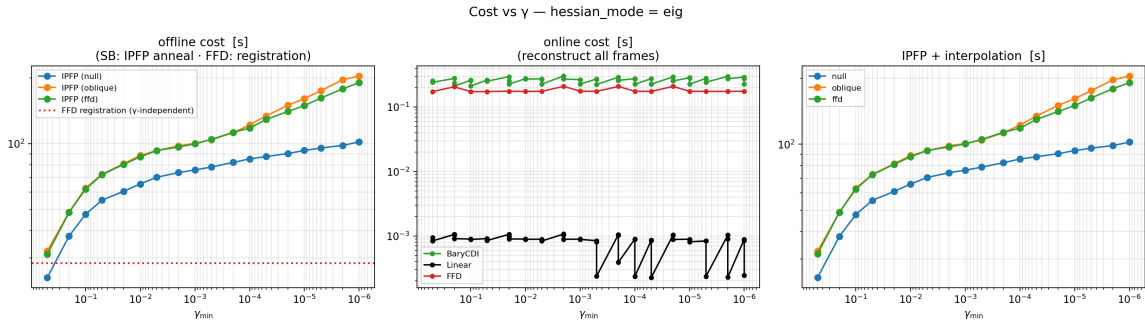
Sweep — Gain over Linear (eig)



Sweep — Cost vs. Accuracy (eig)



Sweep — Cost vs. $\gamma_{\min}$ (eig)



Sweep — IPFP Convergence (eig)

